

Project Summary – Week 4

Data Analytics Track

1. The Specific Problem

As a Data Analytics intern, my role in this project this week is not to solve the entire HealthConnect project, but to answer the central question from a data perspective: what factors are associated with appointment no-shows, and how can appointment data inform a reduction in these no-shows. This work also aims to provide the data understanding and insights needed to support the other tracks of the project (Data Science, Machine Learning Engineering, Generative AI)

2. The Resources

- Python (Pandas, NumPy) within a Jupyter/Colab notebook.
- The appointment dataset provided along with its data dictionary as part of the HealthConnect project

3. My Key Observations from Week 4

- The dataset shows strong overall structural integrity: no duplicates detected, no outliers found in the numerical columns (age, booking lead time, appointment history, distance, waiting time), and no logical inconsistencies between date columns.
- Missing values fall into two categories: those in `reminder_channel` are structurally explained by the absence of a sent reminder; those in `waiting_time_minutes` and `distance_to_clinic_km` (1.8%) are randomly distributed and were handled through imputation mean and median respectively.
- Seven variables relevant to studying no-shows were identified: `booking_lead_days`, `previous_no_shows`, `reminder_sent`, `reminder_channel`, `distance_to_clinic_km`, `appointment_day`, and `appointment_type`.

4. MY Proposed Approach

- 1 . Quantify the no-show rate for each identified variable to calculate the five proposed KPIs
- 2 . Visualise these results to make them easily interpretable
- 3 . Prioritise the most influential factors and translate them into concrete, actionable recommendations for the clinic

5. The Key Considerations that may affect the Project

- Limited appointment history for some patients (low or zero previous_appointments) could limit the reliability of the no-show history KPI.
- Results obtained on this fictional dataset should be interpreted with caution regarding their generalisation to a real clinical context

6. My Proposed Focus for Week 5

- Calculate the five identified KPIs
- Produce clear visualisations for each factor
- Formulate concrete, evidence-based recommendations directly tied to the HealthConnect project's central question on reducing missed appointments